Semiparametric Decomposition of Dengue Transmission Dynamics in Singapore: A Bayesian Renewal Equation Model with Gaussian Process Residuals

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2026-04-17

Abstract

We develop a Bayesian renewal equation model for the weekly dengue effective reproduction number (R_t) in Singapore (2012–2022), decomposing $\log(R_t)$ into additive components for climate covariates and a residual Gaussian process (GP). Using a Hilbert Space GP (HSGP) approximation with a Mat'ern 3/2 kernel, we find that climate (temperature and rainfall) explains less than 1% of R_t variance at weekly resolution, while the residual GP captures 99.4% of temporal variation. Post-hoc analysis of the GP residual against serotype surveillance data reveals that 5 of 7 serotype switch events are associated with elevated transmission (posterior probability > 0.75); cross-correlation shows that the serotype turnover rate leads f_{residual} by $\sim$8 months (peak $r \approx 0.23$, exceeding the $\pm 1.96/\sqrt{n}$ noise band), while Shannon entropy shows no systematic lead–lag relationship. Spectral analysis of f_{residual} reveals multi-scale structure at 26, 51, $\sim$80, and $\sim$187 weeks; the 187-week ($\sim$ 3.6-year) peak is consistent with the multi-annual oscillations that Wearing and Rohani [2006] show can arise in total dengue incidence from transient cross-immunity between serotypes combined with seasonal vector forcing. Cross-kernel comparison demonstrates that conclusions are robust to kernel specification: all four kernels tested (Mat'ern 1/2, 3/2, 5/2, squared exponential) yield statistically indistinguishable fits (ΔELPD < 2). Out-of-sample forecasting reveals a fundamental limitation: the GP provides no predictive skill beyond a random walk baseline at any horizon (CRPS 1.9$\times$, 2.9$\times$, and 6.5$\times$ worse than random walk at 4, 8, and 13-week horizons). Branching process analysis shows that serotype switches create measurable windows of outbreak vulnerability, with 2022 exhibiting the most sustained transmission (18 weeks of $P(\text{sustain}) > 0.5$). A retrospective residual decomposition — comparing full-model weekly expected cases against a climate-only reference while holding infectious pressure fixed — identifies the February 2020 DENV-3 emergence as the only switch where the residual GP accounts for a clearly significant positive case contribution (+4,419; 95% CrI: 2,925 to 6,043). These descriptive associations place serotype dynamics among the plausible drivers of non-climate R_t variation, but a separate model-comparison audit (detailed in the Limitations) shows that explicit SIR-style susceptible-pool reconstructions — aggregate or serotype-specific, across a sensitivity sweep over the initial susceptible fraction — cannot reproduce the GP residual's amplitude or timescale, so the mechanism remains uncertain. The GP framework also has fundamental limitations for causal decomposition and operational forecasting.

1 Introduction

Dengue is the most widespread mosquito-borne viral disease globally, with an estimated 390 million infections annually [Bhatt et al., 2013]. Singapore, despite its advanced public health infrastructure, experiences recurrent dengue epidemics driven by the interplay of climate, vector ecology, population immunity, and the sequential dominance of four dengue serotypes (DENV-1–4) [Rajarethinam et al., 2018]. Understanding the relative contributions of these drivers to transmission intensity is essential for epidemic preparedness and intervention targeting.

The effective reproduction number, R_t—the average number of secondary infections generated by a single case at time t—is the central quantity for real-time epidemic assessment [Cori et al., 2013]. Standard approaches estimate R_t using either the Cori method with a single smoothing window, or flexible Bayesian frameworks such as EpiNow2 [Abbott et al., 2020] that use a Gaussian process (GP) to regularise temporal variation. However, these approaches conflate all sources of R_t variation into a single smooth function, precluding attribution to specific epidemiological drivers.

We address this by building a semiparametric model where $\log(R_t)$ is decomposed into an additive linear predictor for measured covariates (climate) and a residual GP that captures unmeasured temporal structure. GPs are particularly well-suited to this task: they provide principled Bayesian uncertainty quantification over latent functions, allow additive decomposition with parametric covariates, and their kernel hyperparameters (length scale, amplitude) are directly interpretable in epidemiological terms—unlike black-box alternatives such as variational autoencoders or foundation models, which lack this decomposition and interpretability. This framework enables us to ask: conditional on observed climate variation, what residual structure remains in R_t, and does it align with serotype-driven immunity dynamics?

Specifically, this report addresses four questions:

1. How much of weekly R_t variation in Singapore is attributable to climate versus unmeasured drivers?
2. Does the residual GP temporal structure align with serotype replacement events?
3. Are the core findings robust to GP kernel specification?
4. Can the GP framework provide operationally useful forecasts?

We extend this core analysis with branching process analysis, spectral analysis, and retrospective residual decomposition.

2 Methods

2.1 Data Sources

Dengue cases. Weekly notified dengue case counts for Singapore (January 2012 – December 2022) were obtained from the Ministry of Health Weekly Infectious Diseases Bulletin via data.gov.sg. After merging with the weather series and filtering to complete weekly observations, the analysis dataset contains $N = 568$ weeks, of which $N_{\text{model}} = 562$ are fit by the renewal equation after dropping the first 6 weeks as a generation-interval burn-in. Cases include both dengue fever and dengue haemorrhagic fever.

Climate covariates. Weekly mean temperature (°C) and total rainfall (mm) were obtained from the Meteostat API (Singapore Changi Airport, WMO 48698), standardised to unit variance, and lagged by 4 weeks to reflect the time climate conditions take to propagate through *Aedes* vector ecology—egg development, larval and pupal maturation, adult emergence, and biting, approximately two to three weeks—plus the human case-ascertainment delay of intrinsic incubation and onset-to-notification, approximately another two weeks. Note that the extrinsic incubation period is not part of this lag: it is already embedded in the generation interval of the renewal equation.

Serotype proportions. Monthly DENV-1–4 proportions (2013–2022) were derived from NEA sequencing surveillance. Raw monthly proportions are noisy because only a subset of cases is typed each month; the small denominators produce sampling jitter that is not biological signal. Following Finch et al. [2025], we smoothed the raw proportions using a multinomial logistic GAM with cubic regression splines in time, which jointly smooths all four DENV-1–4 proportions while enforcing the simplex constraint that they sum to one and remain non-negative at every time point. Smoothing is a denoising step rather than a modelling assumption: without it, the argmax dominant-serotype label flips noisily from month to month and the dataset contains approximately 23 apparent "switch' ' events, most of which are short sampling artefacts; the GAM smoothing reduces this to 7 genuine replacement events used in all downstream post-hoc analyses. From the smoothed trajectories we computed dominant serotype identity, switch

events, Shannon entropy $H = -\sum_{i=1}^{4} p_i \log p_i$, and the monthly serotype *turnover rate* (the absolute month-to-month change in the smoothed proportion of the currently dominant serotype). Serotype data were used only in post-hoc analyses, not as model covariates, to avoid identifiability problems arising from the fact that serotype proportions are derived from the same case surveillance data used in the outcome.

2.2 Latent Process for R_t

The latent R_t process follows:

$$\log(R_t) = \mu + \beta_{\text{temp}} \cdot \text{temp}_{t-4} + \beta_{\text{rain}} \cdot \text{rain}_{t-4} + f_{\text{residual}}(t) \tag{1}$$

where μ is an intercept, β_{temp} and β_{rain} are regression coefficients for standardised climate covariates at 4-week lag, and $f_{\text{residual}}(t) \sim \mathscr{GP}(0, k_\nu(t,t'))$ is a zero-mean GP with Mat'ern $\nu = 3/2$ kernel:

$$k_{3/2}(t,t') = \alpha^2 \left(1 + \frac{\sqrt{3}\,|t-t'|}{\rho}\right) \exp\left(-\frac{\sqrt{3}\,|t-t'|}{\rho}\right) \tag{2}$$

where α is the marginal standard deviation (controlling amplitude of $\log(R_t)$ excursions) and ρ is the length scale in weeks (controlling the temporal correlation range). The Mat'ern 3/2 produces once-differentiable sample paths—smoother than the Mat'ern 1/2 (Ornstein–Uhlenbeck) but less smooth than the squared exponential (infinitely differentiable). This is the default kernel in EpiNow2 [Abbott et al., 2020] because it permits R_t to change relatively quickly while maintaining continuity.

2.3 HSGP Approximation

Exact GP inference requires $\mathscr{O}(N^3)$ computation for N time points. We use the Hilbert Space basis function approximation [Riutort-Mayol et al., 2023] with $M = 110$ basis functions:

$$f(t) \approx \sum_{j=1}^{M} \beta_j \cdot \phi_j(t), \quad \beta_j \sim \mathscr{N}\left(0, \sqrt{S(\lambda_j)}\right) \tag{3}$$

where $\phi_j(t) = (1/\sqrt{L})\sin(j\pi(t+L)/(2L))$ are Laplacian eigenfunctions on the domain $[-L, L]$ with $L = 1.5 \times (\max(t) - \min(t))/2$, and $S(\omega)$ is the spectral density of the Mat'ern 3/2 kernel:

$$S(\omega) = \alpha^2 \cdot \frac{4(\sqrt{3}/\rho)^3}{((\sqrt{3}/\rho)^2 + \omega^2)^2} \tag{4}$$

This reduces computational complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(MN)$, making inference tractable in Stan [Carpenter et al., 2017] via HMC/NUTS. The choice of $M = 110$ ensures that the highest eigenfrequency exceeds the effective bandwidth of the GP's spectral density at the 5th percentile of the ρ prior.

2.4 Transmission Model

Expected cases follow the discretised renewal equation [Cori et al., 2013]:

$$\lambda_t = R_t \sum_{s=1}^{S} c_{t-s}\, w_s \tag{5}$$

where c_{t-s} are observed cases, w_s is the discretised generation interval PMF ($S = 6$ weeks), and the sum represents the total infectiousness at time t. Rather than fitting a single parametric distribution to published serial interval estimates, the generation interval was constructed by Monte Carlo convolution of its biological components [Chan and Johansson, 2012]: intrinsic incubation period ($T_{\text{IIP}} \sim \text{Gamma}(\text{mean} = 5.9\text{d}, \text{SD} = 1.5\text{d})$), viremic period ($T_{\text{vir}} \sim \text{Uniform}(0, 5\text{d})$), extrinsic incubation period ($T_{\text{EIP}} \sim \text{Gamma}(\text{mean} \approx 10\text{d})$, linearly interpolated from Chan & Johansson (2012)'s anchors of 15 d at 25°C and 6.5 d at 30°C, evaluated at the empirical mean weekly temperature across the 2012–2022 study period (27.91°C), and mosquito biting interval ($T_{\text{bite}} \sim \text{Exponential}(\text{mean} = 2\text{d})$). This yields a mean total GI of 20.4 days (2.92 weeks) with weekly PMF $[0.000, 0.079, 0.504, 0.345, 0.066, 0.007]$; essentially all of the mass falls within 6 weeks, so the first $S = 6$ weeks are treated as burn-in without truncation loss.

2.5 Observation Model

$$c_t \sim \text{NegBin}(\lambda_t, \phi) \tag{6}$$

with $\mathrm{Var}(c_t) = \lambda_t + \lambda_t^2/\phi$, accommodating overdispersion from clustered transmission and spatially heterogeneous reporting. The model assumes constant case ascertainment, which is reasonable for the renewal equation framework since a constant reporting fraction cancels in the R_t calculation [Cori et al., 2013]. Implications of serotype-dependent ascertainment are discussed in Section 4.3.

2.6 Prior Specification

Weakly informative priors encode epidemiological plausibility without strong prescription:

- **Intercept:** $\mu \sim \mathcal{N}(0, 0.5)$, placing the prior median R_t at $\exp(0) = 1$ with 95% interval $[0.37, 2.72]$, spanning the plausible range for endemic dengue.
- **GP length scale:** $\log(\rho) \sim \mathcal{N}(\log(6), 0.5)$, median 6 weeks (95% interval: approximately 2.2 to 16.1 weeks). The relatively short-scale prior reflects the expectation that the residual GP should be able to track the rapid outbreak dynamics not captured by slowly varying climate covariates, while still permitting longer length scales if supported by the data.
- **GP amplitude:** $\log(\alpha) \sim \mathcal{N}(-1.2, 0.5)$, median 0.30 on the $\log(R_t)$ scale, implying typical excursions of ± 0.3 to ± 0.8.
- **Overdispersion:** $\phi \sim$ half-$\mathcal{N}(0, 5)$, tightened from half-$\mathcal{N}(0, 10)$ to push ϕ toward the overdispersed regime; the broader prior let ϕ drift to ~ 47 (near-Poisson), producing PPC intervals too narrow for weekly fluctuations.
- **Covariate coefficients:** $\beta \sim \mathcal{N}(0, 0.5)$, centered at no effect.

2.7 Inference and Model Comparison

Models were fitted using CmdStan 2.38.0 with 4 chains of 1,000 warmup + 1,000 sampling iterations and NUTS defaults except `adapt_delta` $= 0.95$ and `max_treedepth` $= 12$. Convergence was assessed against the thresholds of Vehtari et al. [2021]: split-$\hat{R} < 1.01$, bulk ESS > 400, tail ESS > 400, and zero divergent transitions. Model comparison used approximate leave-one-out cross-validation (LOO-CV) via Pareto-smoothed importance sampling [Vehtari et al., 2017].

2.8 Post-hoc Serotype Analysis

Rather than incorporating serotype data as an in-model covariate, we perform a post-hoc analysis of the residual GP against serotype dynamics. The residual GP $f_{\text{residual}}(t)$ captures R_t variation not explained by climate. If serotype replacement events increase R_t by creating large susceptible pools, then f_{residual} should show a specific temporal profile around switch events: a rise near the switch, followed by gradual decline during stable dominance. All four post-hoc analyses described below operate on the posterior f_{residual} from the single fitted model — no refit, and no additional covariates in Stan.

1. **Switch profiles (event-centred windows).** For each of 7 identified switch events (2013–2022), we extract the posterior distribution of monthly f_{residual} in a 6-month window before and 12 months after the switch. This is the most direct empirical test of the immunity-depletion signature, applied event by event.

2. **Cross-correlation (systematic lead–lag).** We compute cross-correlation functions (CCF) between monthly f_{residual} and two serotype indicators — Shannon entropy and the serotype turnover rate — scanning lags from -12 to $+12$ months. The sign of the peak lag indicates whether the serotype indicator leads or follows residual R_t; the magnitude indicates how strong the systematic relationship is across the full decade. Reported lags use R's standard `ccf` convention: a bar at lag k is the correlation between $f_{\text{residual},t+k}$ and the indicator at month t, so a positive peak lag means the indicator leads.

3. **Spectral analysis (theory-driven cycling test).** We compute the power spectrum of the posterior median f_{residual} via a Hanning-windowed FFT and propagate uncertainty across posterior draws to obtain a credible band on the spectrum. This is an aggregate test against the theoretical prediction from multi-strain SIR models that cross-immunity between four co-circulating serotypes produces multi-annual oscillations at specific timescales [Wearing and Rohani, 2006].

4. **Per-switch residual case contribution (retrospective decomposition).** For each switch, we compare full-model weekly expected cases $\lambda_t = R_t \cdot \text{ip}_t$ against a climate-only reference $\lambda_t^{\text{ref}} = R_t^{\text{ref}} \cdot \text{ip}_t$ with $\log R_t^{\text{ref}} = \mu + f_{\text{climate}}$, holding the observed infectious pressure ip_t fixed, and cumulate the per-week difference over the 6 months following each switch. This translates the log-scale residual signal into a per-event case count. As described in

Section 2.9, this is a retrospective week-by-week decomposition rather than a forward-simulated counterfactual history.

The four methods provide complementary lenses on the same residual series: switch profiles handle individual events, the CCF handles systematic lead–lag, spectral analysis tests theory-driven cycling predictions, and the per-switch decomposition converts the signal into case numbers. All are descriptive: with only 7 switch events, the sample size precludes formal hypothesis testing.

2.9 Extensions

Climate lag sensitivity. The baseline 4-week lag reflects *Aedes* vector ecology (egg-to-biting adult, ~2–3 weeks) plus intrinsic incubation and onset-to-notification (~2 weeks). To verify that the headline result is not an artifact of this choice, the core model was refitted at lags of 2, 3, 4, and 6 weeks, using the same standardisation constants for comparability. All four fits sampled cleanly (zero divergences).

Cross-kernel comparison. The core model (Equation 1) was refitted with Mat'ern 1/2, 3/2, 5/2, and squared exponential kernels to assess sensitivity of conclusions to smoothness assumptions. The only change is the spectral density function $S(\omega)$ in Equation 3; all other model components remain identical.

Out-of-sample forecasting. We evaluate the GP's predictive skill via a rolling-origin design on a 26-week late-2022 holdout window. Four forecast origins are placed one per month from July through September 2022 (marked by red diamonds in Figure 10); at each origin the model is refitted on all data up to and including that week, and 4-, 8-, and 13-week-ahead forecasts are drawn from the renewal-equation posterior. We emphasise upfront that with only four origins, all within a single late-2022 low-incidence window, this is an illustrative experiment rather than a comprehensive benchmark. Probabilistic forecasts are scored with the continuous ranked probability score (CRPS),

$$\mathrm{CRPS}(F_{\mathrm{pred}}, y_{\mathrm{obs}}) = \int_{-\infty}^{\infty} \left(F_{\mathrm{pred}}(x) - \mathbf{1}\{x \geq y_{\mathrm{obs}}\}\right)^2 dx, \tag{7}$$

which is the integrated squared gap between the predictive CDF F_{pred} and the step function located at the observation y_{obs} (visualised in Figure 10 right panel). CRPS is in the same units

as the observation (cases), is strictly proper (both calibration and sharpness count), and lower is better. We compare the GP against two baselines: a random walk on cases (persistence — the minimum bar for any time-series forecast) and a trivial $R_t = 1$ endemic reference. Skill against a baseline is computed as $1 - \mathrm{CRPS}_{\mathrm{GP}}/\mathrm{CRPS}_{\mathrm{baseline}}$, so positive skill means the GP wins. Serotype Shannon entropy at 43-week lag was additionally tested as a candidate leading indicator.

Branching process analysis: from R_t to outbreak risk. The estimated R_t is an *average* number of secondary cases per primary case. A complementary question is: if a single case arrives in week t, what is the probability that the transmission chain it seeds *sustains* rather than dying out by chance? This is a stochastic question that the mean R_t alone cannot answer — even when $R_t > 1$, small introductions may go extinct by bad luck; when R_t is just below 1, a lineage can still persist for several generations. We answer it with a Galton–Watson branching process applied week-by-week to the posterior R_t trajectory.

Offspring distribution. Each infected case is assumed to generate a random number of secondary infections Y drawn from a negative binomial distribution, $Y \sim \mathrm{NB}(\text{mean} = R_t,\ \text{dispersion} = k)$. The negative binomial allows the variance to exceed the mean and thus accommodates the superspreading phenomenon documented for directly-transmitted pathogens [Lloyd-Smith et al., 2005], which a Poisson offspring distribution cannot capture.

Extinction equation. Let q denote the probability that a lineage started by a single infected individual eventually dies out. The classical Galton–Watson result gives q as the smallest non-negative fixed point of $q = G(q)$, where $G(s) = \mathbb{E}[s^Y]$ is the probability generating function (PGF) of the offspring distribution. For the negative binomial,

$$G(s) = \left(1 + \tfrac{R_t}{k}(1-s)\right)^{-k}.$$

When $R_t \leq 1$ the only root in $[0,1]$ is $q = 1$ (extinction is certain), and when $R_t > 1$ a second root appears in the open interval $(0,1)$, which is the true extinction probability. We solve this fixed-point equation numerically with `uniroot` on $h(s) = G(s) - s$, with fixed-point iteration as a fallback.

Outbreak-sustaining probability. We define the outbreak-sustaining probability as

$$P(\text{sustain}) = 1 - q.$$

This quantity is highly non-linear near $R_t = 1$: small shifts in R_t near the critical threshold translate into large shifts in sustained-chain risk, which makes it a more sensitive summary of vulnerability than R_t itself. We evaluate $P(\text{sustain})$ *per posterior draw* of (R_t, ϕ) at every modelled week (thinning to 1,000 draws for runtime) and summarise as the median with 95% credible interval, so uncertainty in the HSGP-derived R_t and in the dispersion flows through the branching-process calculation coherently.

Proxy for offspring dispersion. Because the person-to-person offspring dispersion k cannot be identified from case counts alone, we substitute the fitted observation-model dispersion, $k \leftarrow \phi$, as a pragmatic proxy. Strictly, ϕ describes week-to-week overdispersion in observed case counts relative to the renewal-equation mean rather than person-to-person variance in secondary infections, so the absolute $P(\text{sustain})$ values should be read with this substitution in mind. The qualitative shape of the $P(\text{sustain})$ trajectory is driven almost entirely by R_t. Further limitations of applying a single-type Galton–Watson model to dengue — homogeneous mixing, infinite susceptibles, the absence of an explicit mosquito vector stage, and the omission of serotype structure and antibody-dependent enhancement — are discussed in Section 4.3.

Spectral analysis. The posterior mean f_{residual} was analysed via Lomb–Scargle periodogram to identify dominant periodicities and compare them against theoretical predictions from multi-serotype dynamics.

Residual decomposition. The residual GP's contribution to expected cases was computed by comparing the full-model weekly expected cases $\lambda_t = R_t \cdot \text{ip}_t$ against a climate-only reference $\lambda_t^{\text{ref}} = R_t^{\text{ref}} \cdot \text{ip}_t$ with $\log R_t^{\text{ref}} = \mu + f_{\text{climate}}$ (residual GP set to zero). Both use the *same* observed infectious pressure ip_t. This is a retrospective week-by-week decomposition, *not* a forward-simulated counterfactual history: we do not iterate the renewal equation under R_t^{ref} from $t = 1$ onward, and the residual GP additionally absorbs all non-parametric-climate structure, so the resulting per-switch numbers should be read as descriptive accounting rather than causal attribution.

3 Results

3.1 Core Model: Climate Explains Less Than 1% of R_t Variance

The posterior predictive distribution tracks the timing and magnitude of all major epidemic peaks (Figure 3), but pointwise intervals under-cover weekly fluctuations — a limitation of the negative binomial observation layer discussed in Section 4.3. MCMC convergence diagnostics are uniformly clean (Figure 1): zero divergent transitions, zero max-tree-depth hits, split-$\hat{R} <$ 1.01 for every scalar parameter and for every f_{residual} component (max $\hat{R} = 1.003$ across all 562 components), bulk ESS $>$ 1,900 for all scalar parameters (with most in the thousands), and minimum bulk ESS of 3,647 across the f_{residual} components. No reparameterisation was required, because the HSGP construction uses basis coefficients that are iid standard normal by design and already in the right geometry for HMC. Trace plots for the four core hyperparameters (μ, α, ρ, ϕ) show well-mixed, overlapping chains with no trends. Any remaining concerns about the decomposition are therefore scientific rather than computational.

Climate effects on R_t are negligible at weekly resolution. A 1-SD increase in temperature multiplies R_t by 0.983 (95% CrI: 0.962, 1.011); a 1-SD increase in rainfall multiplies R_t by 1.000 (95% CrI: 0.982, 1.020). Neither effect is distinguishable from 1.0. Variance decomposition confirms that climate covariates explain only 0.6% (median) of R_t variation, with the residual GP absorbing 99.4% (Figure 2).

This result requires careful interpretation. The GP's dominance is partly a structural consequence of model specification: with a short estimated length scale ($\hat{\rho} = 2.83$ weeks; 95% CrI: 2.06, 3.66), the GP can track week-to-week outbreak dynamics that climate covariates—operating on seasonal timescales—cannot explain. The finding that climate is negligible *at weekly resolution* does not preclude climate being an important driver at seasonal or interannual timescales; rather, weekly R_t variation is dominated by faster-acting processes.

3.2 Serotype Associations with the GP Residual

Post-hoc analysis of the GP residual against serotype surveillance data reveals consistent associations between serotype switch events and elevated transmission (Figure 4). Of 7 identified switch events (2013–2022), 5 show posterior probability $>$ 0.75 of elevated f_{residual} in the 0–3

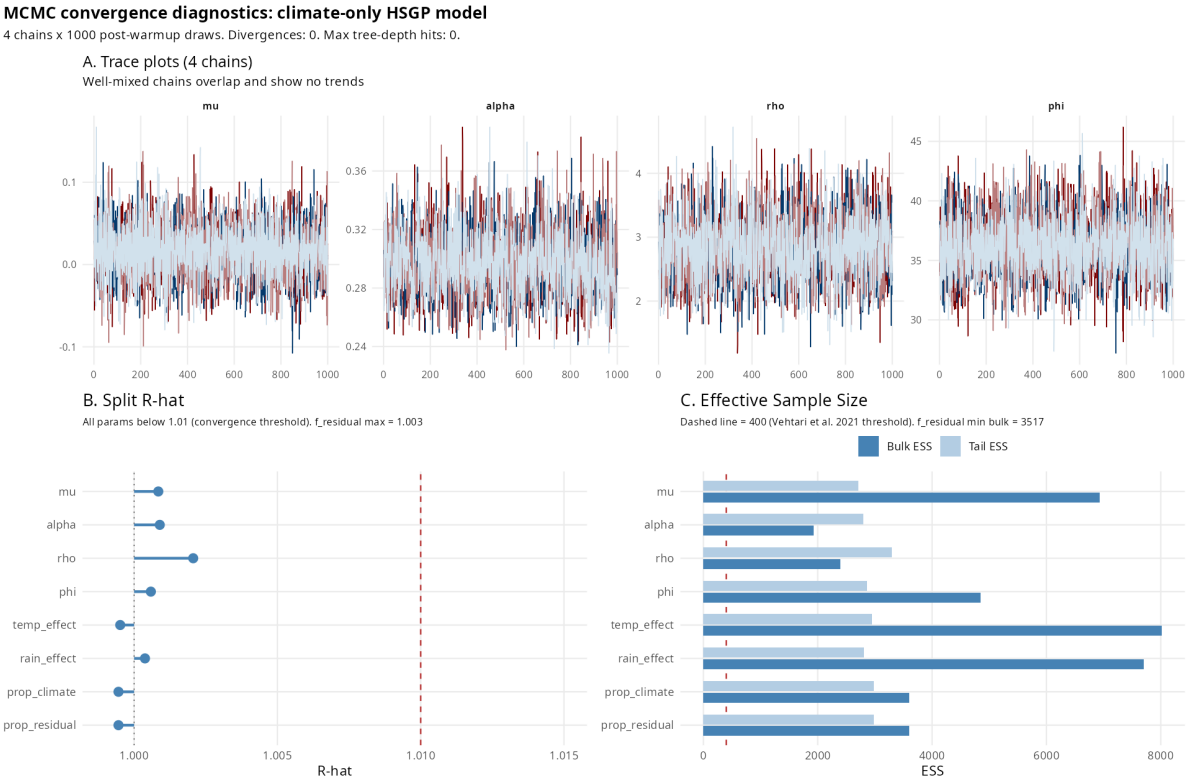

Figure 1: MCMC convergence diagnostics for the climate-only HSGP model (4 chains $\times$ 1,000 post-warmup draws; HMC/NUTS with `adapt_delta` $= 0.95$). **Panel A:** Trace plots for the core hyperparameters μ, α (GP amplitude), ρ (GP length scale in weeks), and ϕ (negative binomial dispersion), showing well-mixed overlapping chains. **Panel B:** Split-$\hat{R}$ for each scalar parameter with the $\hat{R} < 1.01$ convergence threshold [Vehtari et al., 2021] shown as a red dashed line; annotation reports the maximum $\hat{R}$ across the 562 f_{residual} components. **Panel C:** Bulk and tail effective sample sizes for each scalar parameter; the red dashed line marks the 400 threshold [Vehtari et al., 2021]. Zero divergent transitions and zero max-tree-depth hits across all chains.

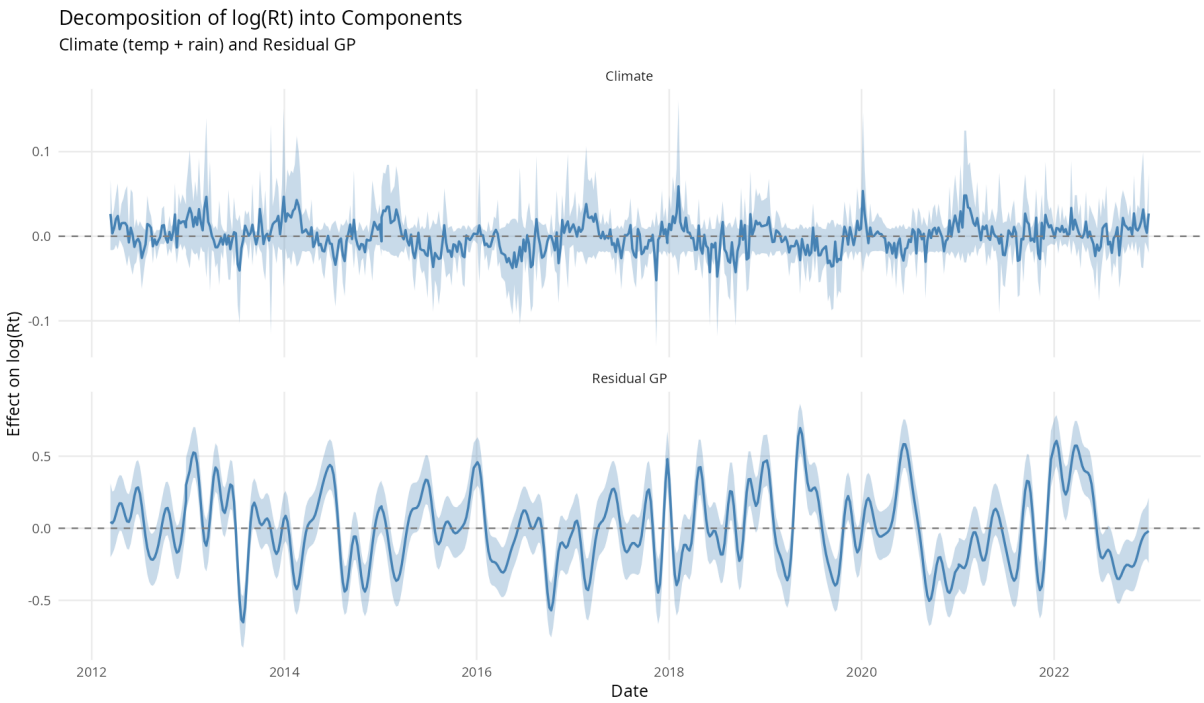

Figure 2: Decomposition of $\log(R_t)$ into climate and residual GP components. The residual GP (orange) captures nearly all temporal structure in R_t, while climate covariates (blue) contribute negligibly. Shaded bands show 95% credible intervals. The GP length scale of approximately 3 weeks allows tracking of individual outbreak dynamics.

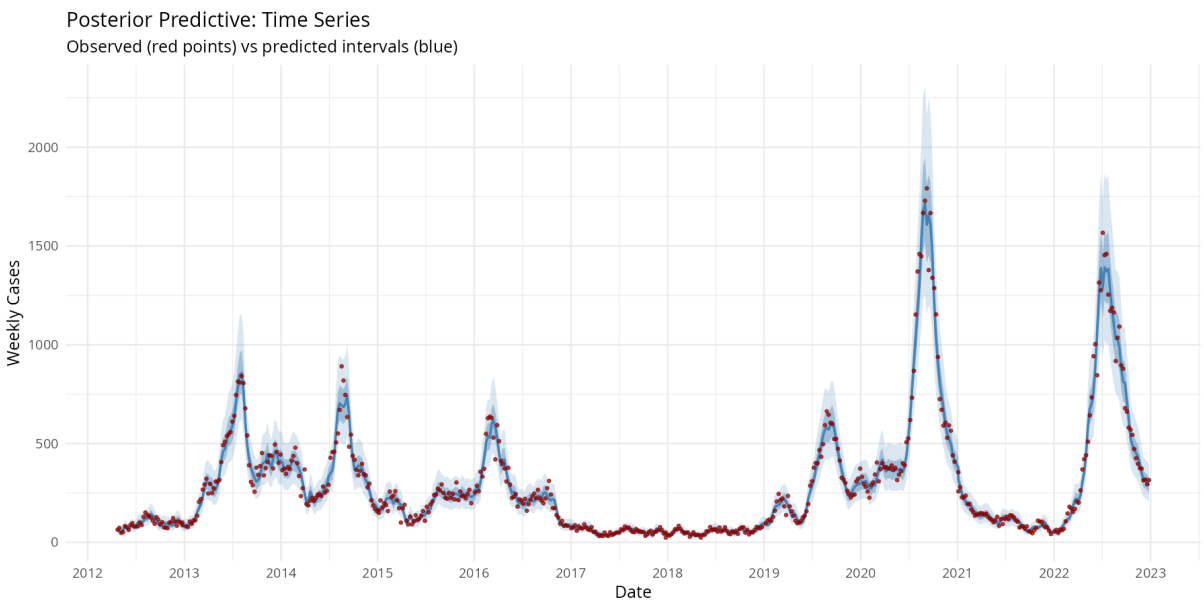

Figure 3: Posterior predictive check: observed weekly dengue cases (black points) against posterior predictive 95% intervals (shaded blue). The model captures the timing and magnitude of major epidemic peaks (2013, 2019, 2020, 2022); pointwise intervals under-cover weekly fluctuations (discussed in Section 4.3).

months post-switch.

The DENV-2→DENV-3 transition in February 2020 produces the strongest signal ($P = 1.0$; 20–35% R_t increase beyond climate prediction). The mechanism behind this elevation remains uncertain: a separate model-comparison audit (Section 4.3) shows that the GP residual cannot be reproduced by explicit susceptible-pool reconstructions (aggregate or serotype-aware) under simple SIR accounting, so "naive susceptible pool" readings of the DENV-3 elevation should be treated with caution.

Cross-correlation analysis tests two serotype indicators against monthly f_{residual}: Shannon entropy, and the serotype *turnover rate* (absolute month-to-month change in the dominant serotype's smoothed proportion). The turnover rate shows a clear peak at lag +8 months ($r \approx 0.23$), with lag +9 immediately adjacent ($r \approx 0.19$); only lag +8 exceeds the $\pm 1.96/\sqrt{n}$ noise band (Figure 6). Shannon entropy, by contrast, stays within the noise band at every lag — so the actionable signal lies specifically in the *rate of change* of dominant-serotype proportions, not in diversity per se. A ∼8-month lead between turnover and residual R_t elevation is consistent with the timescale of susceptible-pool replenishment following a serotype replacement, and is potentially actionable for seasonal risk assessment — though not for tactical weekly forecasting (see Section 3.5), where the relationship is too weak to translate into skill.

These associations are descriptive with $n = 7$ switch events, providing no meaningful sample size for formal hypothesis testing. The evidence is pattern-based: the temporal profiles match predictions from immunity-depletion theory in both direction (positive post-switch) and timing (aligned with observed serotype transitions), but cannot exclude alternative explanations.

3.3 Climate Lag Sensitivity

Refitting the model at alternative climate lags of 2, 3, 4 (baseline), and 6 weeks confirms that the headline variance decomposition is not an artifact of the lag choice (Figure 7). The climate variance share remains below 3% at every lag (lag 2: 2.6% [0.9, 5.3]; lag 3: 1.3% [0.3, 3.1]; lag 4: 0.4% [0.03, 1.5]; lag 6: 1.9% [0.6, 4.1]), and the residual GP captures at least 97% in every case. The temperature coefficient flips sign across lags (R_t multipliers of 1.049 at lag 2, 1.025 at lag 3, 0.986 at lag 4, and 0.964 at lag 6), which itself is diagnostic: no single weekly lag identifies a robust, stable climate signal, consistent with the interpretation that climate acts primarily on slower timescales. GP hyperparameters ($\hat{\rho}$, $\hat{\alpha}$) are essentially unchanged across

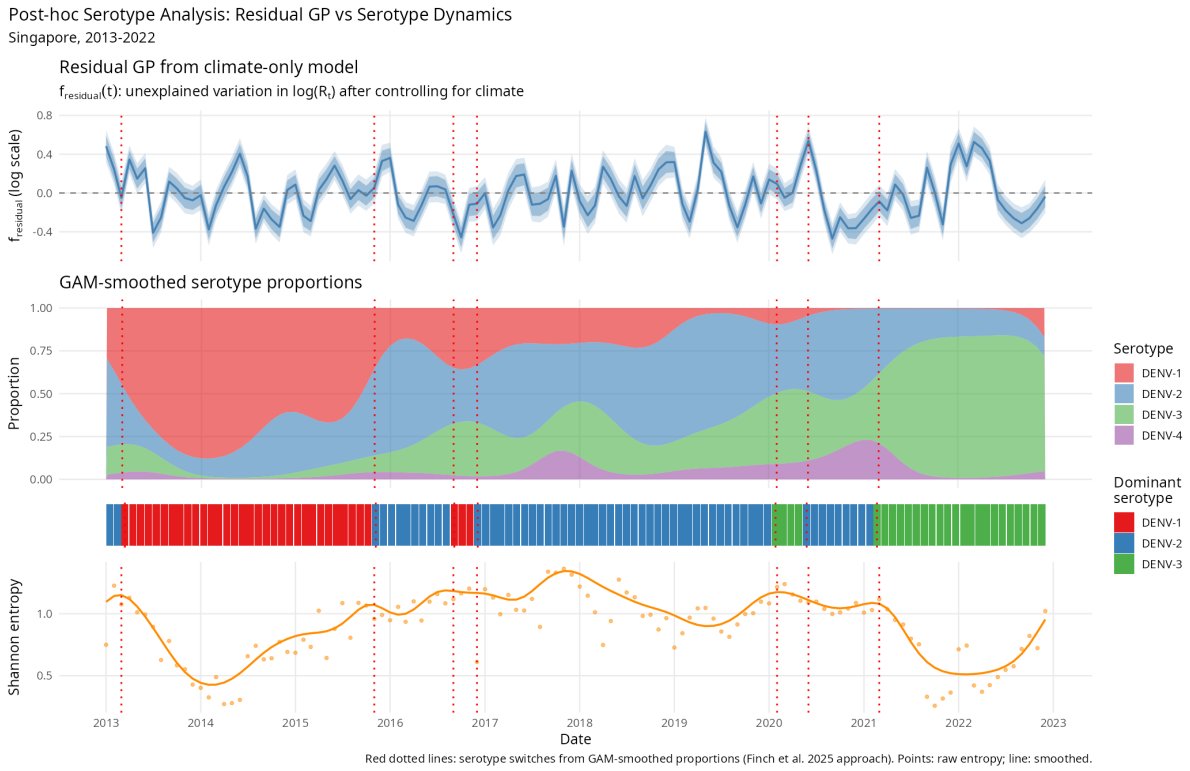

Figure 4: Three-panel serotype analysis showing temporal alignment between the GP residual and serotype dynamics. **Top:** Posterior median f_{residual} with 80% (dark) and 95% (light) credible bands. **Middle:** GAM-smoothed serotype proportions (DENV-1 red, DENV-2 blue, DENV-3 green, DENV-4 purple). **Bottom:** Shannon entropy computed from smoothed proportions. Vertical dashed lines mark serotype switch events. Five of 7 switches coincide with positive f_{residual} excursions.

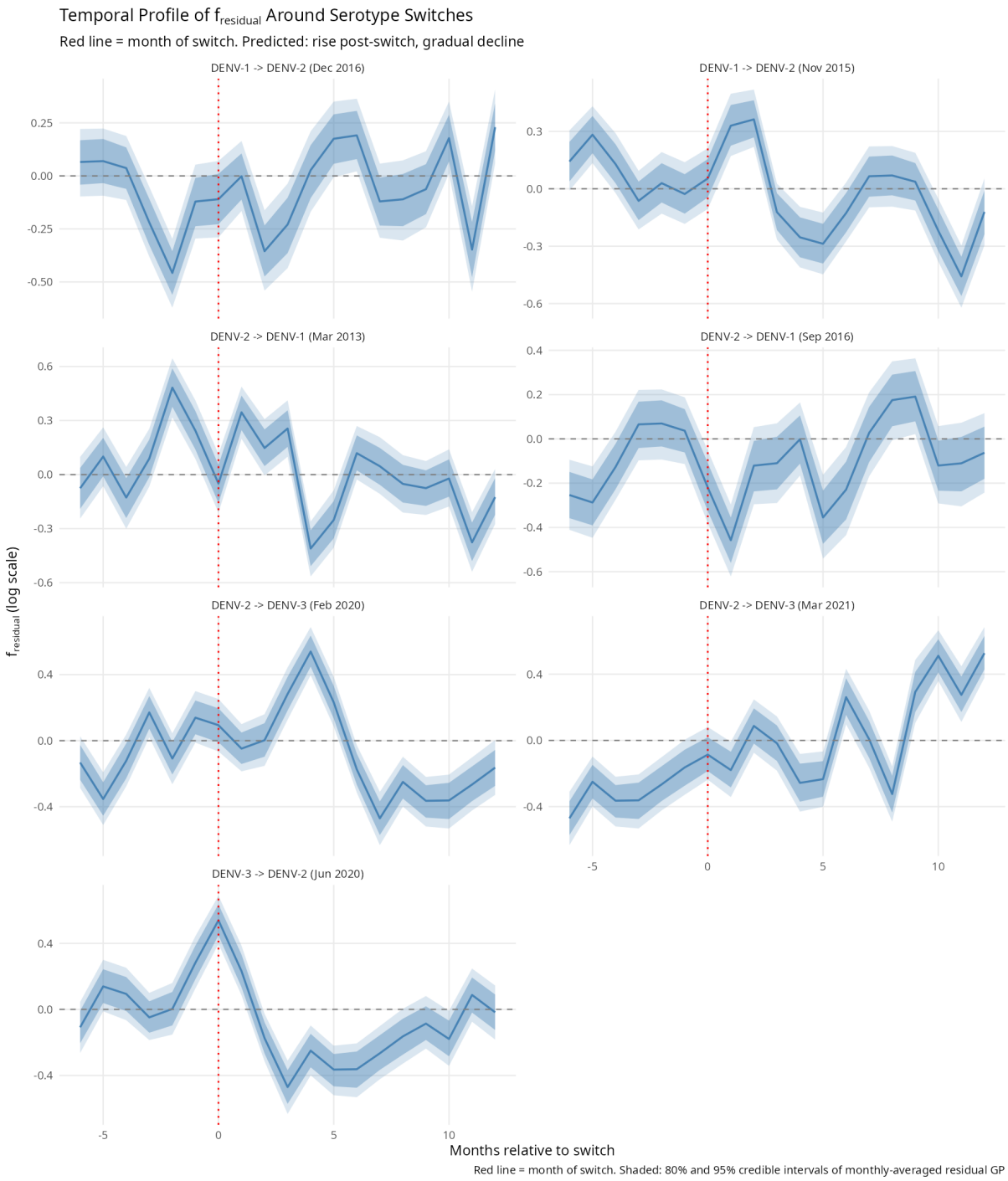

Figure 5: Temporal profiles of f_{residual} centred on each serotype switch event (time zero). Each panel shows one switch, with posterior median (line) and 95% credible band (shading). The predicted immunity-depletion pattern—rise at switch, gradual decline during stable dominance—is visible in most events, most clearly for the 2020 DENV-3 emergence.

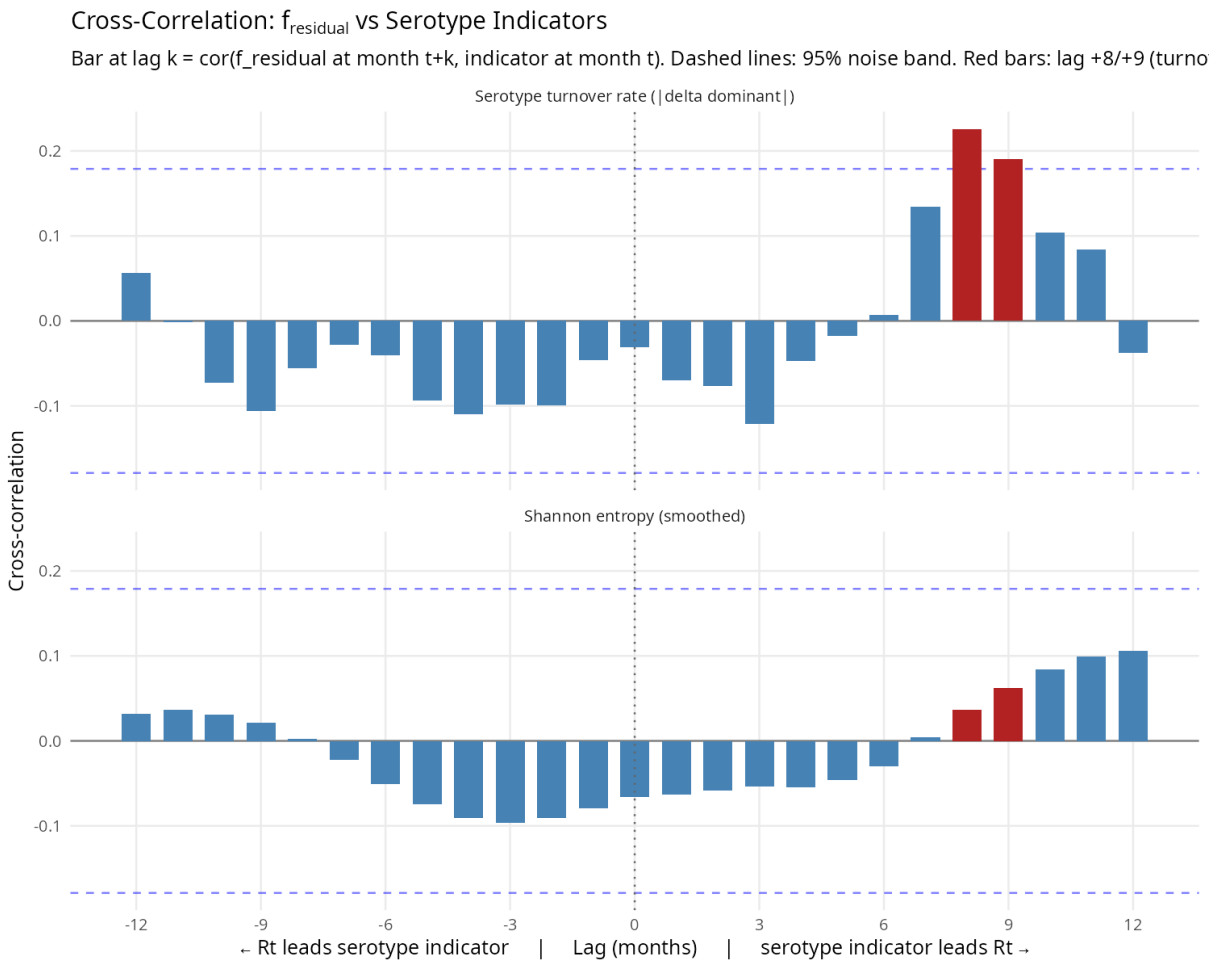

Figure 6: Cross-correlation function between monthly f_{residual} and two serotype indicators: Shannon entropy (left) and serotype turnover rate (right; absolute month-to-month change in dominant serotype proportion). Positive lag k means the indicator at month t predicts f_{residual} at month $t+k$. The turnover rate peaks at lag $+8$ months ($r \approx 0.23$, highlighted in red; the only bar exceeding the $\pm 1.96/\sqrt{n}$ noise band), implying that turnover precedes residual R_t elevation by approximately 8 months. Shannon entropy remains within the noise band at every lag, so the lead–lag signal lies specifically in the rate of change of dominant-serotype proportions rather than in diversity.

all lag specifications. All four fits sampled cleanly with zero divergences.

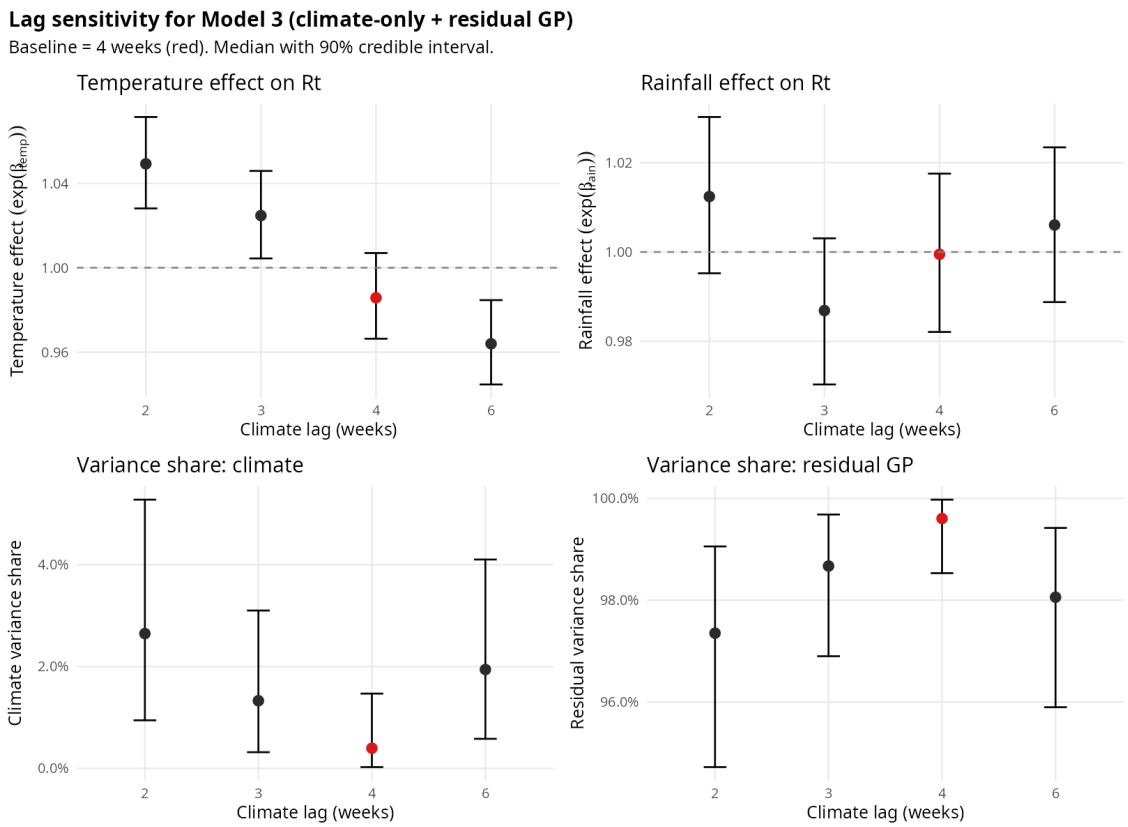

Figure 7: Climate lag sensitivity. Top row: temperature and rainfall multiplicative effects on R_t (posterior median with 90% credible interval) at lags 2, 3, 4, and 6 weeks; dashed line marks the null. Bottom row: posterior variance share of the climate component and the residual Gaussian process. Baseline lag (4 weeks) highlighted in red. Climate share remains below 3% at every lag; temperature direction flips across lags, indicating no single weekly lag identifies a stable climate signal.

3.4 Cross-Kernel Comparison: Conclusions Are Robust

All four kernels produce statistically indistinguishable fits, with $\Delta\text{ELPD} < 2$ and overlapping standard errors (Figure 8). The roughest kernel (Mat'ern 1/2) is marginally preferred (ELPD $= -2839.4$), consistent with dengue R_t exhibiting abrupt changes. The Mat'ern 3/2 is second ($\Delta\text{ELPD} = -0.8$, SE $= 0.5$), followed by the squared exponential ($\Delta\text{ELPD} = -1.6$, SE $= 0.7$) and Mat'ern 5/2 ($\Delta\text{ELPD} = -1.8$, SE $= 0.6$).

MCMC diagnostics are clean across all kernels: zero divergences, $\hat{R} \leq 1.01$, minimum bulk ESS > 805. Crucially, the serotype detectability pattern is preserved across kernels, with all four detecting the same 5/7 elevated-probability switches (Figure 9). This robustness confirms that

the choice of Mat'ern 3/2 is defensible but not critical—all scientific conclusions hold regardless of kernel specification.

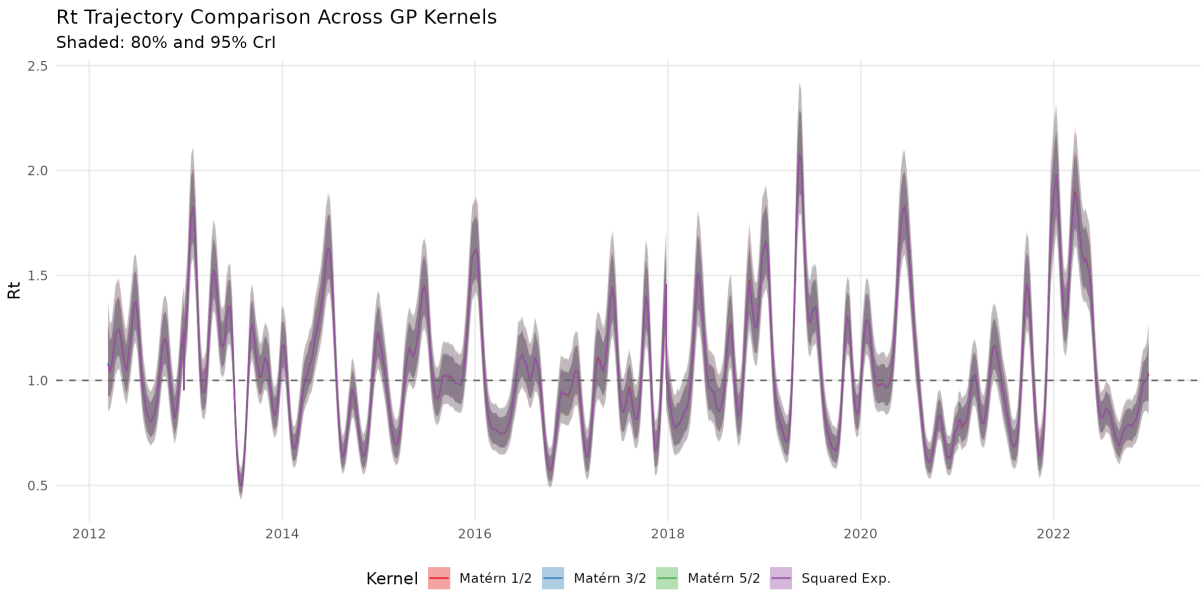

Figure 8: Overlaid R_t trajectories from all four kernel specifications: Mat'ern 1/2 (blue), Mat'ern 3/2 (orange), Mat'ern 5/2 (green), and squared exponential (red). Posterior medians are nearly identical; differences are smaller than credible interval widths. ΔELPD values are shown in the legend.

3.5 Forecasting: The GP Provides No Predictive Skill

The rolling-origin design is shown in Figure 10 and what CRPS measures, against one example prediction-observation pair, is shown in Figure 11. The GP never beats a random walk baseline at any forecast horizon (Figure 12): mean CRPS is 1.9×, 2.9×, and 6.5× worse than random walk at the 4-, 8-, and 13-week horizons, corresponding to skill scores of −7.6, −2.3, and −10.5 respectively (the GP is *worse* than persistence). The GP additionally loses to the trivial $R_t = 1$ baseline at 8 and 13 weeks. Calibration is conservative: 95% forecast intervals achieve 100% empirical coverage, indicating overly wide prediction bands rather than good calibration.

Both the GP and the $R_t = 1$ baseline converge to similar predictions at medium horizons because the GP's posterior R_t collapses toward the prior mean within $\hat{\rho} \approx 3$ weeks of any forecast origin (Figure 13). Serotype Shannon entropy at 43-week lag, tested as a candidate leading indicator, does not improve forecasts at any horizon — unsurprisingly, given that the CCF analysis showed entropy itself stays within the noise band at every lag, and that the actionable signal lies in

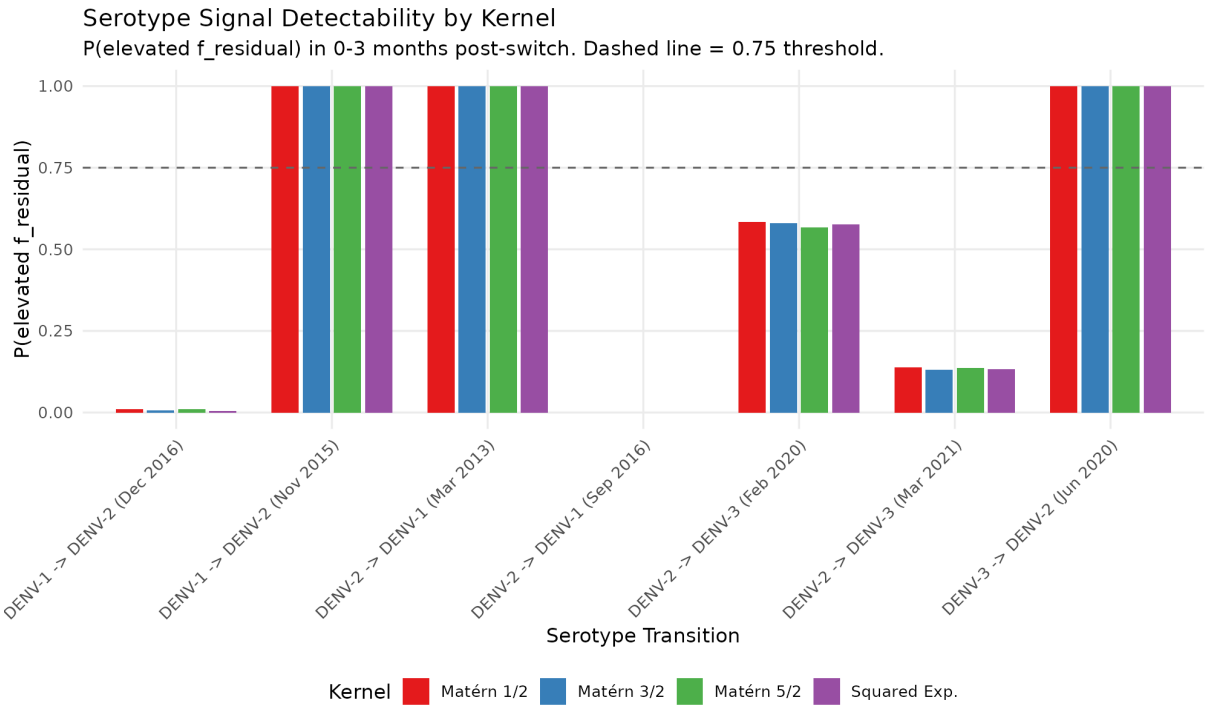

Figure 9: Serotype switch detectability across kernel specifications. Each bar shows $P(\text{elevated } f_{\text{residual}})$ in the 0–3 months post-switch. All four kernels identify the same 5 of 7 switches (horizontal dashed line at $P = 0.75$), confirming that serotype–R_t associations are not an artefact of the Mat'ern 3/2 assumption.

the turnover rate with an ∼8-month lead which is too long a timescale for tactical weekly forecasting.

This is a fundamental, not incidental, limitation. The GP is a smoothing prior that regularises temporal variation retrospectively; it contains no mechanistic forward model and therefore provides no information about future trajectory beyond mean reversion. For operational dengue forecasting, models with explicit vector dynamics and immunity tracking are likely necessary.

3.6 Branching Process: Serotype Switches Create Vulnerability Windows

The posterior R_t and overdispersion parameter yield a strong anticorrelation between R_t and extinction probability ($\rho_{\text{Spearman}} = -0.94$; Figure 14). Five of 7 serotype switches show increased outbreak sustaining probability ($P(\text{sustain})$) after the switch, with a mean increase of 6 percentage points.

The year 2022 stands out with the most sustained transmission: 18 weeks of $P(\text{sustain}) > 0.5$ and mean $R_t = 1.17$. This coincides with a period of DENV-3 circulation, consistent with

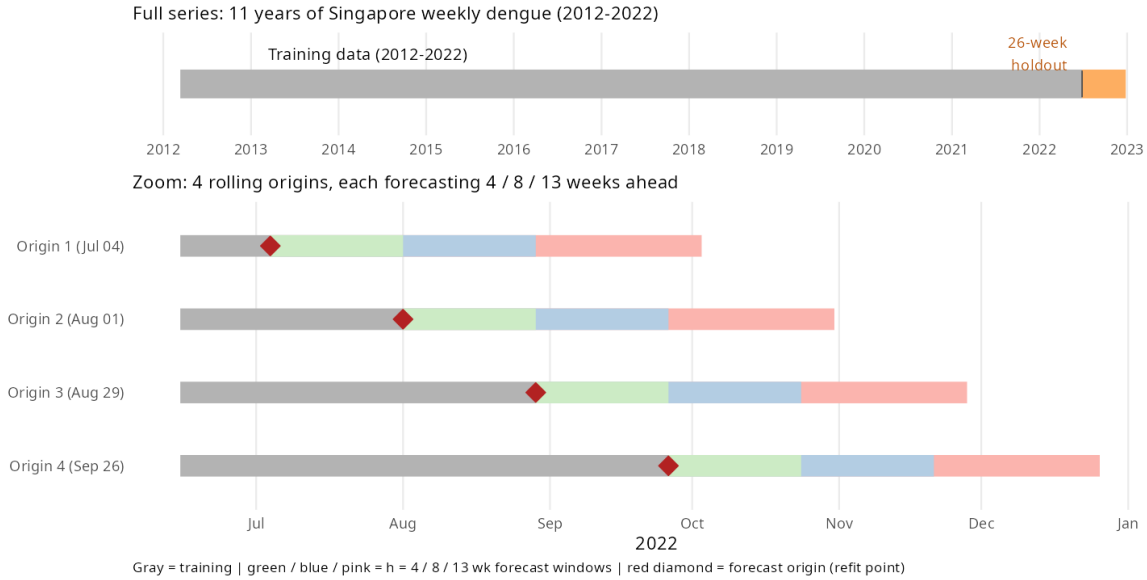

Figure 10: Rolling-origin design. The top strip shows the full 11-year Singapore series with the 26-week late-2022 holdout window highlighted in orange; the bottom panel zooms in and shows four refit cutoffs, one every four weeks from July through September 2022 (red diamonds). From each cutoff the model forecasts 4, 8, and 13 weeks ahead (green, blue, and pink segments, respectively).

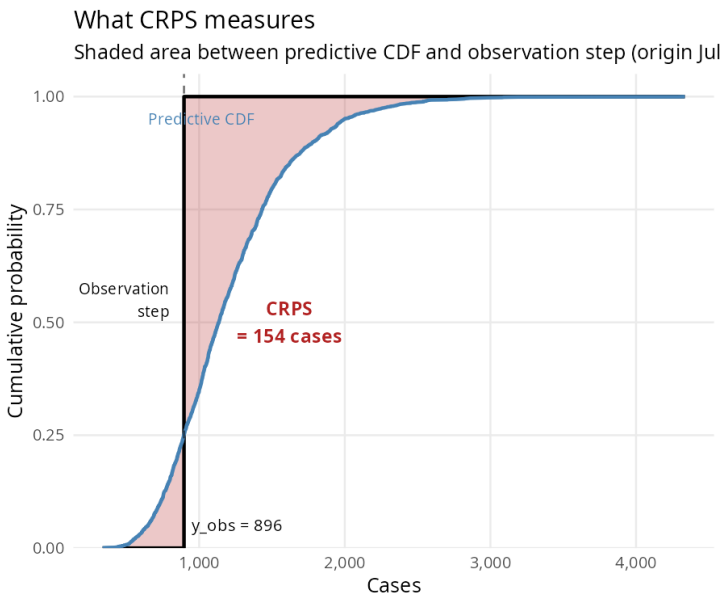

Figure 11: Visualisation of what the continuous ranked probability score (CRPS) measures, for one real forecast point (origin 2022-07-04, horizon 4 weeks). CRPS is the integrated squared gap between the predictive CDF (blue) and the step function at the observed value $y_{\text{obs}} = 896$ (black); the shaded red region is the CRPS. Smaller area means the prediction concentrated probability mass closer to the observed value.

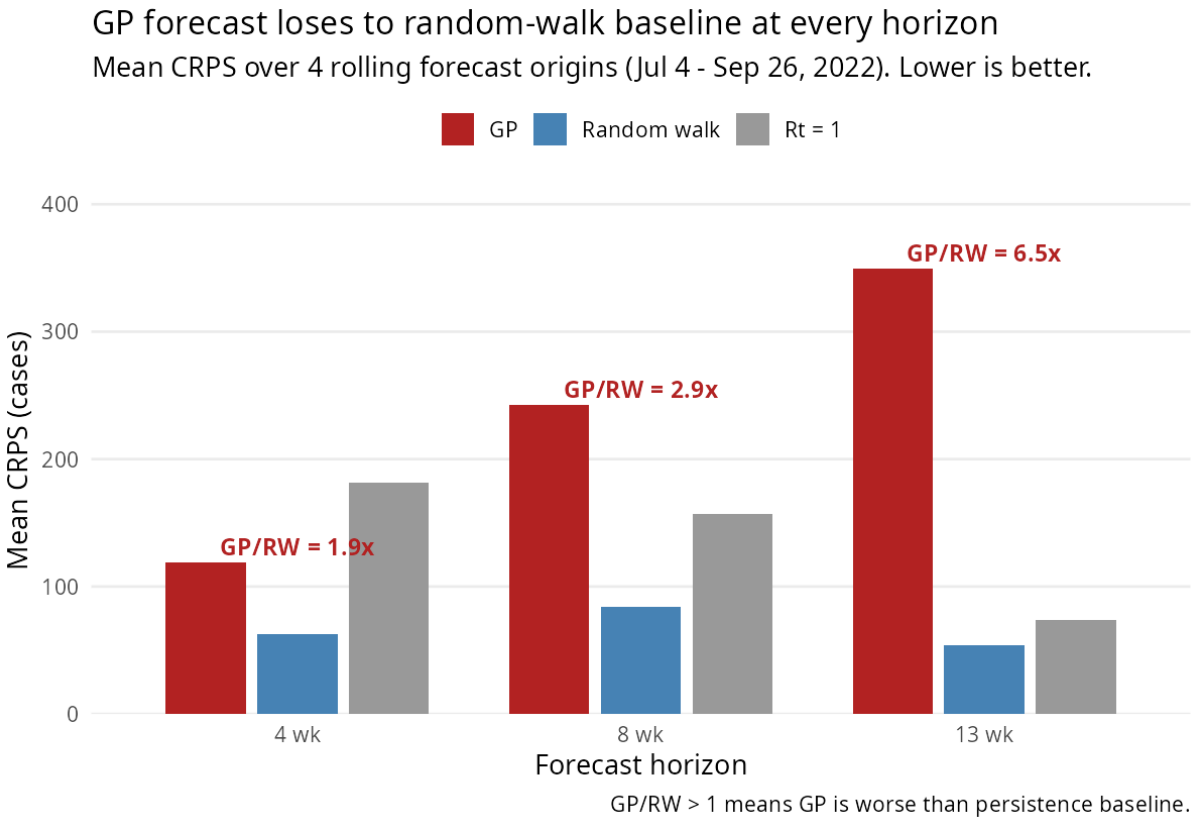

Figure 12: CRPS comparison across forecast horizons: GP (red) versus random walk baseline (blue) and $R_t = 1$ baseline (grey), averaged over the four forecast origins. Lower CRPS indicates better probabilistic forecast accuracy. Annotations show the GP/RW ratio at each horizon; the GP is 1.9 to 6.5 times worse than persistence at every horizon tested.

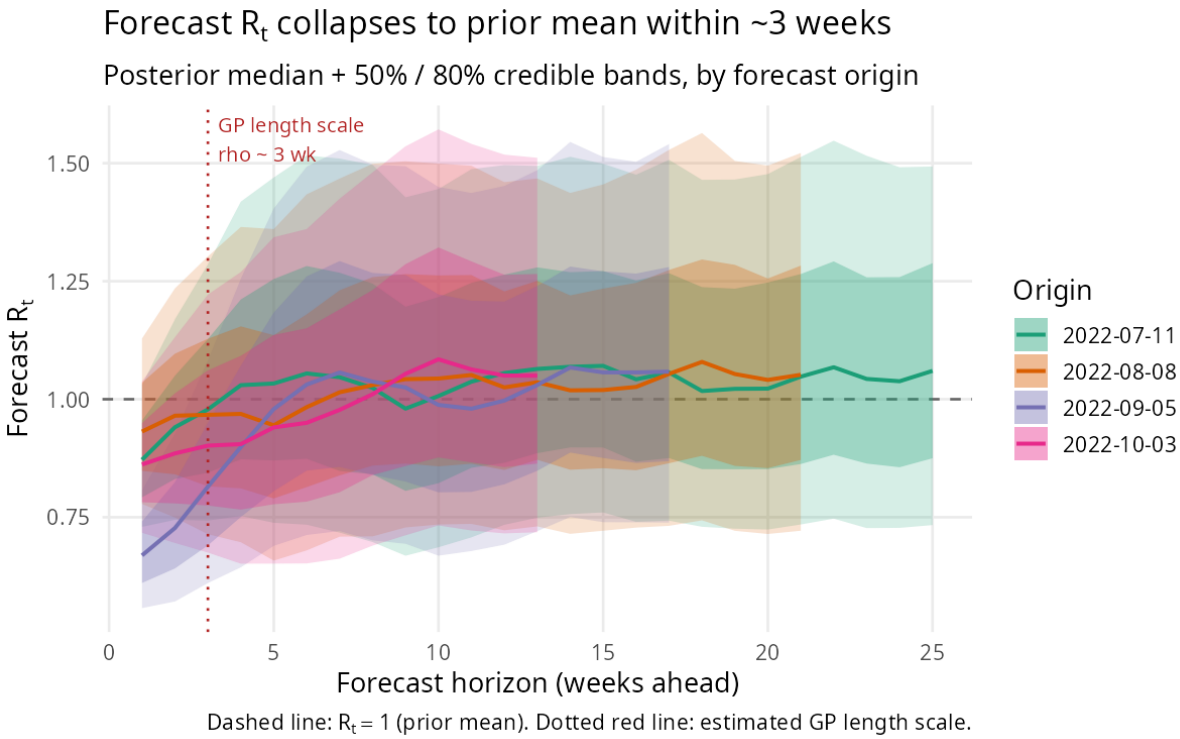

Figure 13: Forecast R_t mean reversion. Posterior median R_t (coloured lines) and 50%/80% credible bands, by week-ahead, for each of the four forecast origins. A dotted vertical guide marks the estimated GP length scale $\hat{\rho} \approx 3$ weeks. The posterior inflates to the prior within about three weeks of any origin: this is the mechanism for the failure in Figure 12.

serotype-driven immunity depletion sustaining elevated transmission.

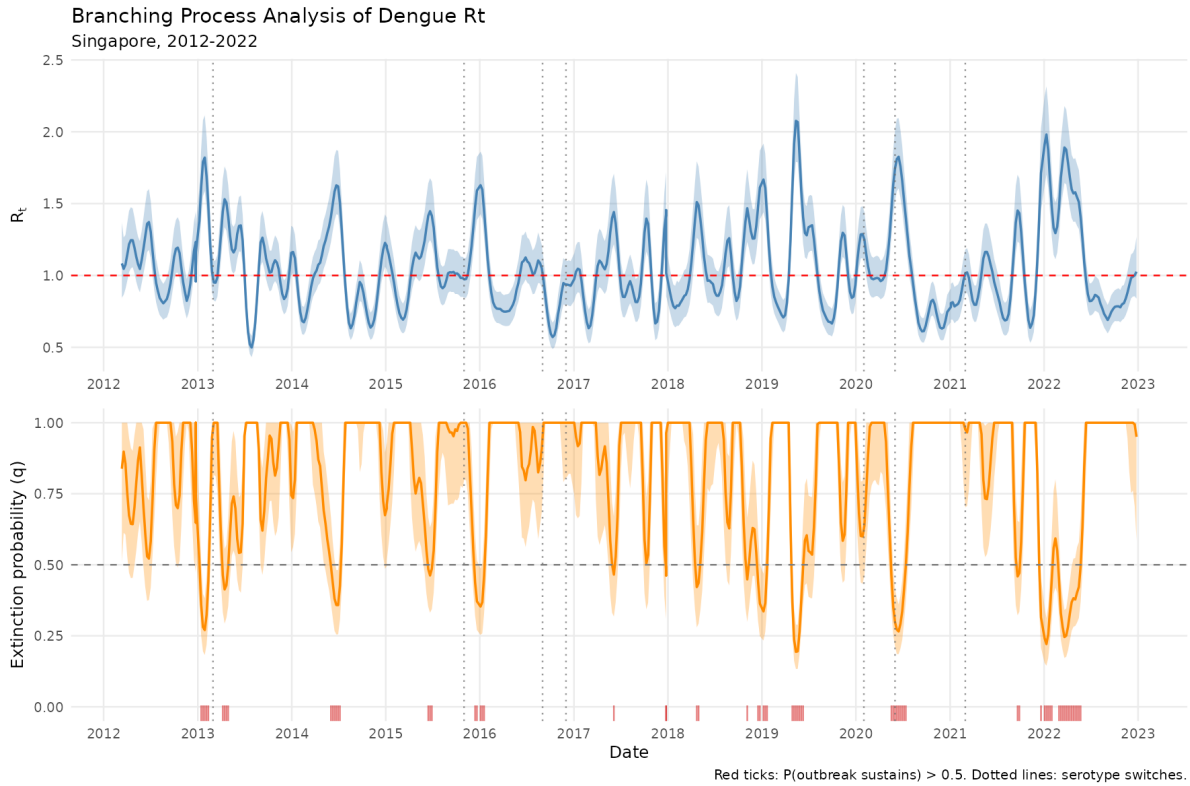

Figure 14: Branching process analysis. **Top:** Posterior median R_t trajectory with 95% credible band. **Bottom:** Time-varying outbreak sustaining probability $P(\text{sustain}) = 1 - q$, where q is the extinction probability under the negative binomial offspring distribution. Vertical dashed lines mark serotype switch events. The strong anticorrelation ($\rho_{\text{Spearman}} = -0.94$) between R_t and extinction probability confirms that small changes in R_t translate to large changes in outbreak risk.

3.7 Spectral Analysis: Multi-Scale Temporal Structure

Method-wise, we compute the power spectrum of the posterior median f_{residual} via a Hanning-windowed FFT and propagate uncertainty across posterior draws to obtain a credible band on the spectrum. The four dominant peaks (top 4 by spectral power, with minimum log-period separation enforced to prevent near-duplicate labels) are:

- **26 weeks ($\sim$ 6 months):** Strongest peak, consistent with the biannual dengue epidemic pattern driven by Singapore's two monsoon seasons (Northeast: Dec–Mar, Southwest: Jun–Sep).
- **51 weeks ($\sim$ 1 year):** Residual annual seasonality not absorbed by the weekly temperature and rainfall covariates, possibly reflecting non-climatic seasonal drivers (school

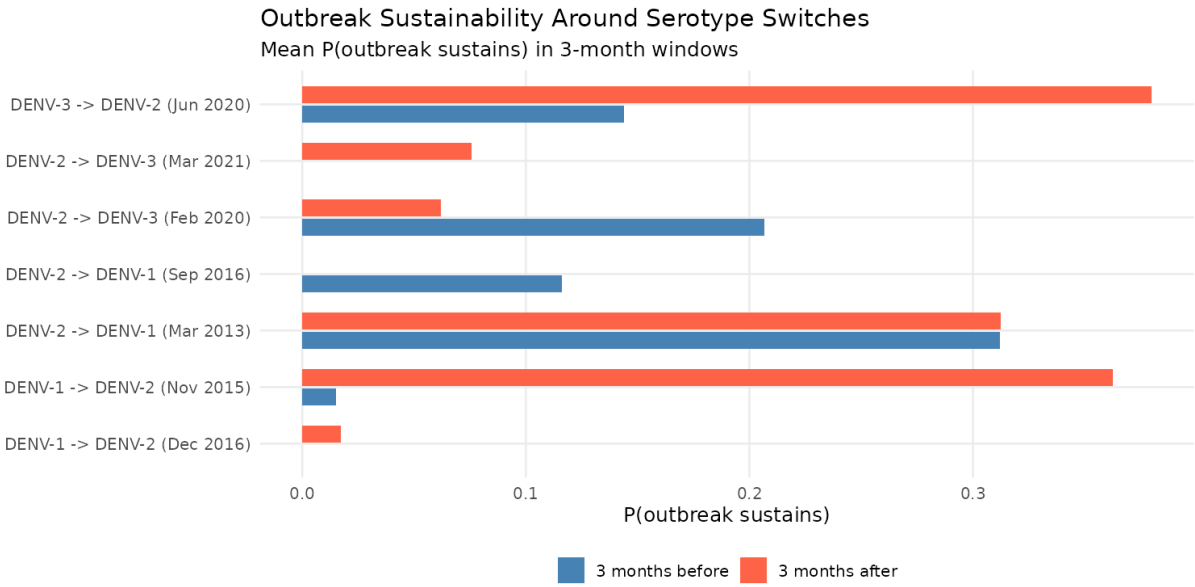

Figure 15: Outbreak sustaining probability before (grey) versus after (coloured) each serotype switch, computed as the mean $P(\text{sustain})$ in 3-month windows. Five of 7 switches show increased vulnerability post-switch (mean increase: $+6$ percentage points), consistent with the immunity-depletion mechanism.

terms, travel patterns, vector control campaign timing). Its presence in the *residual* after climate removal indicates that the climate covariates at weekly resolution do not fully capture all seasonal drivers of dengue transmission.

- **80 weeks ($\sim$ 1.5 years):** Multi-annual structure beyond annual seasonality.
- **187 weeks ($\sim$ 3.6 years):** Consistent with the multi-annual oscillations that Wearing and Rohani [2006] show can arise in total dengue incidence from transient heterologous cross-immunity between serotypes combined with seasonal vector forcing. We do not claim this is evidence of "four-serotype rotation" or of individual-serotype cycling, both of which would require a much longer observation window than 11 years to resolve.

The 187-week peak is the most theoretically substantive finding: its location is consistent with the predicted period for multi-serotype oscillations driven by transient cross-immunity. We emphasise that the 11-year observation window admits fewer than three complete cycles at this timescale, so the evidence is suggestive rather than conclusive. The 80-week peak is less cleanly aligned with any specific theoretical prediction and should be read as descriptive multi-annual structure.

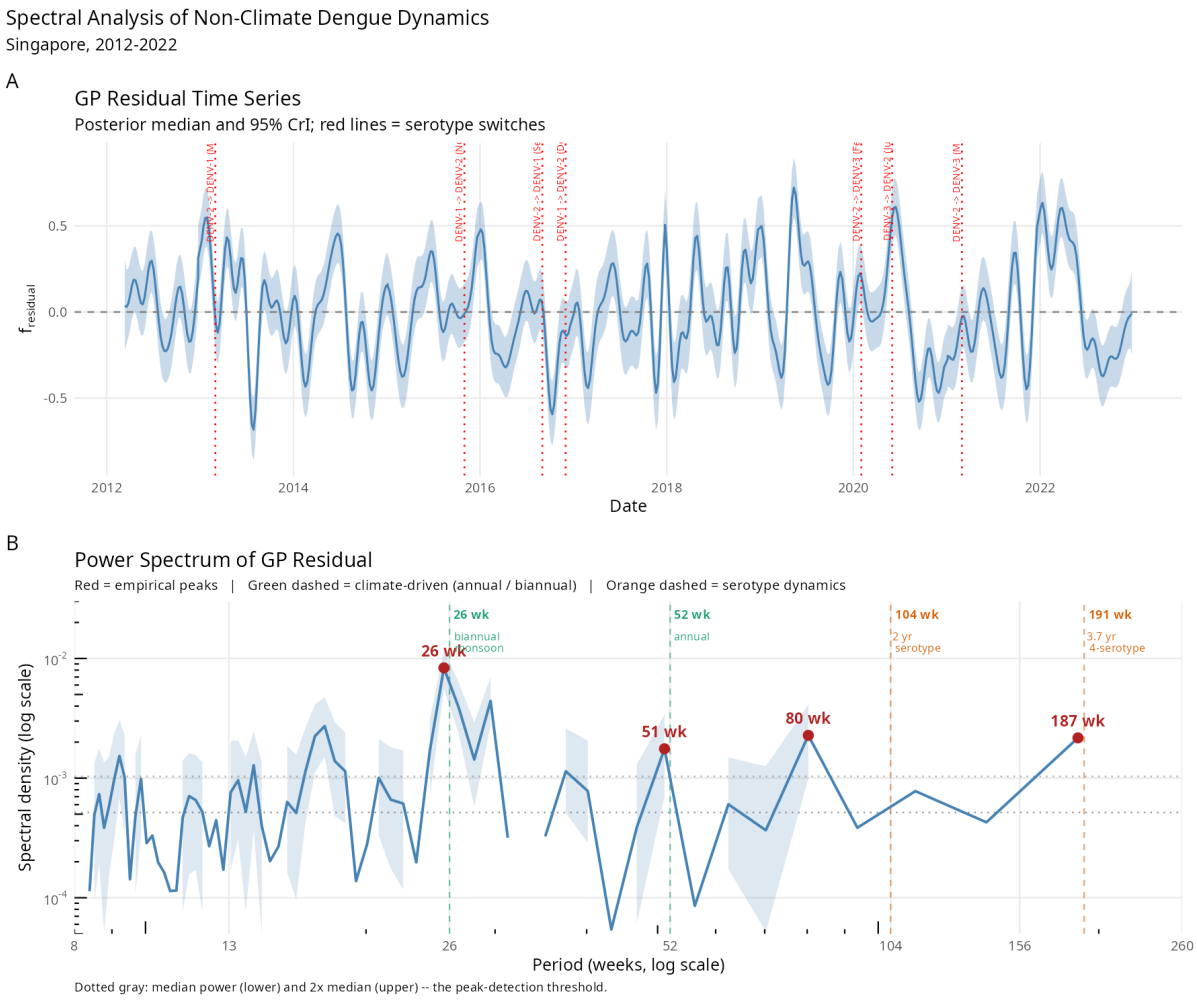

Figure 16: Spectral analysis of f_{residual}: (top) posterior median f_{residual} time series with switch events marked by red dotted lines, (bottom) Hanning-windowed FFT power spectrum with 95% credible interval shaded, and the four dominant peaks annotated in red. Peak labels show only the top 4 by spectral power with a minimum log-period separation to prevent overlap. The spectrum reveals structure at the biannual monsoon cycle (26 wk), residual annual seasonality (51 wk), and multi-annual ($\sim$1.5 yr and $\sim$ 3.6 yr) timescales. The $\sim$ 3.6 yr peak is consistent with multi-annual oscillations in total dengue incidence driven by transient cross-immunity plus seasonal forcing [Wearing and Rohani, 2006].

3.8 Residual Case Contribution Decomposition

The February 2020 DENV-3 emergence is the only serotype switch where the residual GP produces a clearly significant positive case contribution: +4,419 cases (95% CrI: 2,925 to 6,043) over the 6 months post-switch (Figure 17). Over the full decade, positive and negative contributions approximately cancel (net median: −763 cases; 95% CrI: −12,129 to +10,556).

Most switches show contributions indistinguishable from zero, suggesting that only *novel serotype introductions*—not rotations between familiar serotypes—produce a detectable residual signal at the national level. The September 2016 DENV-2→DENV-1 and March 2021 DENV-2→DENV-3 switches show marginally significant *negative* contributions (−693 and −296 respectively), potentially reflecting post-epidemic troughs. As noted in the Methods, these numbers are a retrospective week-by-week decomposition against observed infectious pressure and should be read as accounting, not as a causal claim about cases prevented.

4 Discussion

4.1 Summary of Findings

This study demonstrates that a semiparametric Bayesian framework—combining a renewal equation with an additive GP residual—can effectively decompose dengue R_t into measured and unmeasured components. Climate is genuinely negligible at weekly resolution (0.6% of variance). The residual GP captures multi-scale temporal structure at biannual (26-week), annual (51-week), multi-annual (∼80-week), and multi-year (∼187-week, ∼ 3.6-year) timescales; the 3.6-year peak is consistent with the multi-annual oscillations in total dengue incidence that Wearing and Rohani [2006] show can arise from transient cross-immunity combined with seasonal vector forcing. Five of 7 serotype switches associate with elevated R_t, and the branching process analysis shows these create measurable windows of outbreak vulnerability.

However, the framework has clear limitations. The GP absorbs any smooth temporal variation, making causal attribution to serotype dynamics inherently ambiguous. The GP provides no forecasting skill: in the rolling-origin experiment it is 1.9×, 2.9×, and 6.5× worse than a random walk baseline in CRPS at 4-, 8-, and 13-week horizons, and its posterior median R_t regresses to the prior mean within approximately $\hat{\rho} \approx 3$ weeks of the forecast origin.

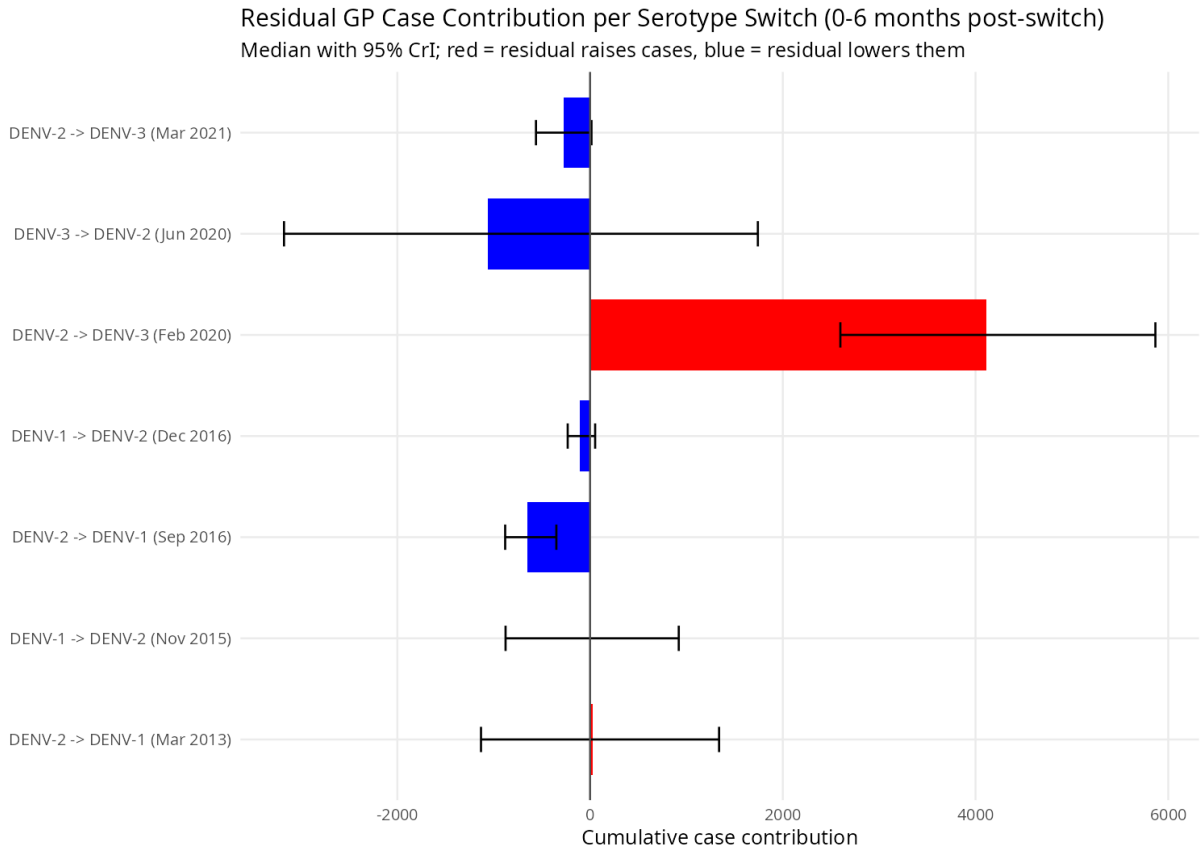

Figure 17: Residual GP contribution to expected cases per serotype switch event, computed as cumulative full-model minus climate-only-reference expected cases over 6 months post-switch. Both trajectories are evaluated against the *same* observed infectious pressure; this is a retrospective week-by-week decomposition, not a forward-simulated counterfactual history. Points show posterior median; error bars show 95% CrI. Only the February 2020 DENV-3 emergence (red) produces a significant positive contribution.

4.2 Comparison with Existing Literature

Our finding that climate is a weak predictor of weekly dengue R_t is consistent with Finch et al. [2025], who found that climate explains only a modest fraction of dengue dynamics in Singapore after accounting for serotype competition. The serotype-immunity mechanism we describe is well-established theoretically [Reich et al., 2013] but has rarely been quantified through GP decomposition of the R_t time series.

The spectral analysis reveals a ~ 3.6-year peak in the f_{residual} power spectrum, consistent with the multi-annual oscillations in total dengue incidence that Wearing and Rohani [2006] show can be generated by transient heterologous cross-immunity combined with seasonal vector forcing. We stop short of claiming individual-serotype cycling (which Reich et al. [2013] and others place on ∼8–10 year timescales and which the 11-year observation window cannot resolve). The 26-week dominant periodicity likely reflects the bimodal seasonal pattern driven by Singapore's two monsoon seasons.

4.3 Limitations

1. **Climate decomposition is scale-dependent.** The finding that climate explains $<1% of R_t variance applies specifically at weekly resolution. Climate likely operates through seasonal modulation of vector populations on longer timescales. A model with explicit seasonal kernels or monthly aggregation might attribute more variance to climate.

2. **GP dominance is partly structural, and the serotype interpretation is not supported by a direct model comparison.** With an estimated length scale of ~3 weeks, the GP can track any smooth temporal variation, whether or not it originates from serotype dynamics. The 99.4% variance share is a consequence of the model giving the GP sufficient flexibility to absorb nearly all signal—it does not constitute evidence that the residual is *caused by* serotype dynamics. A separate methodological audit (`code/14_*`) refits the model with $\log S(t)$ added as a covariate alongside the GP residual, across a sensitivity sweep over the initial susceptible fraction $S_0 \in \{0.50, 0.75, 0.95\}$ and an additional historical configuration with per-serotype $S_j(0)$ values derived from observed 2013–2014 serotype proportions ($S_{D1}(0) \approx 0.65$ to $S_{D4}(0) \approx 0.99$). Across all 7 fits in the sweep, the GP amplitude $\hat{\alpha}$ is essentially unchanged (0.305 to 0.308 vs the 0.307 baseline), β_S

credible intervals include zero in every configuration, and no LOO ELPD difference vs the baseline exceeds its own standard error. In other words, *explicit* SIR-style susceptible structure cannot absorb the GP residual under any defensible parameterisation, which weakens the strong "the residual is serotype-driven immunity depletion" reading of the post-hoc associations. Plausible alternatives the framework cannot distinguish include importation pulses correlated with serotype transitions, spatial heterogeneity in immunity that aggregates poorly to the national level, vector or behavioural dynamics that happen to co-vary with serotype shifts, or individual-level immune complexity (homotypic vs heterotypic protection, cross-immunity, antibody-dependent enhancement) that simple SIR pool accounting cannot represent. A further methodological caveat applies to any such comparison: $\log S(t)$ is constructed from cumulative cases, which is also the model outcome, so a positive ELPD bump would carry built-in feedback and should not be taken as mechanistic evidence even if it were observed.

3. **Serotype associations are descriptive with low statistical power.** With only 7 switch events in 10 years, formal hypothesis testing is not meaningful.

4. **Case ascertainment is assumed constant.** While a constant reporting fraction cancels in the renewal equation [Cori et al., 2013], serotype-dependent ascertainment could bias the post-hoc serotype analysis. Tan et al. [2019] estimate infection-to-notification ratios ranging from 3:1 to 16:1 depending on serotype and period. Time-varying ascertainment is not identifiable from case data alone.

5. **Forecasting failure is fundamental.** The GP's inability to beat a random walk baseline reflects the inherent limitation of a smoothing prior that contains no mechanistic forward model, not a tuning problem. The rolling-origin forecasting experiment is also severely limited in scope: only four origins in a single late-2022 low-incidence window, so the result is illustrative rather than a comprehensive benchmark.

6. **Spatial aggregation.** The model treats Singapore as a single spatial unit, potentially masking heterogeneity in neighbourhood-level transmission driven by local vector densities and demographic structure.

7. **Generation interval is time-invariant.** The discretised generation interval is fixed for the whole study period: we use a single PMF computed at the empirical mean weekly

temperature (27.91°C, EIP $\approx$ 10 d, GI mean 20.4 d), rather than a time-varying PMF recomputed each week from that week's temperature. Weekly temperature ranges from 24.8 to 30.6°C in the data, which over the Chan & Johansson (2012) interpolation corresponds to an EIP range of roughly 6 to 15 days; the constant-GI approximation therefore under-represents variability at the week-to-week scale. Sensitivity analysis across point-estimate temperature-dependent EIP scenarios shows qualitatively stable results, but jointly estimating the generation interval under temperature forcing could improve inference.

8. **Single time series.** The 11-year observation window provides limited statistical power for characterising multi-year cycles (e.g., the 3.6-year peak in the spectral analysis corresponds to fewer than 3 complete cycles).

9. **Galton–Watson assumptions are unrealistic for dengue.** The branching-process analysis adopts a single-type Galton–Watson model with negative binomial offspring, which is a stylised risk lens rather than a faithful description of dengue transmission. Several of its core assumptions are violated: (i) *homogeneous mixing* — dengue transmission is spatially clustered around mosquito breeding sites, households, and workplaces, so effective offspring varies dramatically by location; (ii) *independent offspring* — cases cluster because the same mosquitoes bite nearby people, so offspring within a lineage are correlated; (iii) *infinite susceptible pool* — in a local outbreak the susceptibles in a neighbourhood are quickly depleted, producing endogenous saturation the branching process cannot capture; (iv) *no cross-immunity or antibody-dependent enhancement (ADE)* — a secondary heterologous infection can be *more* transmissible or severe than a primary one, which a single-type model collapses away; (v) *no age or mobility structure*. We also use the fitted observation-model dispersion ϕ as a proxy for the offspring dispersion k, which is not a direct estimate of person-to-person variance in secondary infections. The reported $P(\text{sustain})$ values should therefore be read as an order-of-magnitude indication of how close each week is to the critical point $R_t = 1$, not as a calibrated forecast of outbreak outcomes; a faithful risk model would require a stochastic SEIR or individual-based model with explicit vector dynamics, age groups, serotype structure, and spatial heterogeneity.

4.4 Future Directions

The most important extension is **a spatially explicit, multi-serotype susceptible-infected-recovered (SIR) model fitted at the cluster level using NEA dengue cluster surveillance**. Spatial disaggregation and serotype structure are not two separable extensions but a single interlocking one, and the present analysis directly motivates both. The susceptible-pool audit reported in Section 4.3 — a 60-cell grid sensitivity sweep (`code/13_susceptible_sensitivity.R`) and a proper model-comparison test with $\log S(t)$ as a covariate (`code/14_*`, across a sweep of initial susceptible fractions including a per-serotype configuration derived from observed early proportions) — shows that smooth national-level susceptible compartments cannot reproduce the GP residual's amplitude or timescale under any defensible parameterisation. Mechanistically generating residual variation of the size we observe therefore requires *both* serotype-specific susceptible compartments at a finer spatial grain (so local sub-population pools can be exhausted and replenished much faster than national averages) and spatial heterogeneity (so that sub-population pockets of naive susceptibles, visible at the cluster level, are not averaged out). Serotype structure alone at the national scale would still miss the spatial pockets; spatial disaggregation alone, without serotype compartments, would still be unable to explain why some switches (e.g., the February 2020 DENV-3 emergence) produce thousands of attributable residual cases while others (e.g., the 2016 DENV-1/DENV-2 rotations) are essentially silent. The two are load-bearing for each other.

A spatially-aware multi-strain SIR also addresses the framework's structural forecasting failure. The present GP framework regresses to its prior within $\hat{\rho} \approx 3$ weeks of any forecast origin and is $1.9\times$ to $6.5\times$ worse than a random walk baseline at 4–13 week horizons, because it is a smoothing prior with no mechanistic forward model. A multi-strain SIR, by contrast, integrates the renewal equation forward under explicit immunity dynamics, providing the operational forecasting capability the descriptive GP structurally lacks. Singapore is unusually well-positioned for this extension: NEA publishes dengue case data at the cluster level, the Wolbachia release sites (Yishun, Tampines, Choa Chu Kang) provide neighbourhood-level natural experiments, and the serotype surveillance series spans the full study period. The aggregation to the national scale in the present work was a modelling choice driven by the desire to keep the GP decomposition tractable, not a data constraint.

One further smaller extension is worth flagging. **Benchmarking against deep generative baselines** (variational autoencoders, time-series transformers, foundation models for epidemiology) on matched feature sets would quantify how much of the GP residual structure is recoverable from raw case data alone; a preliminary single-architecture VAE pilot did not provide a fair test of this question and is left as future work.

4.5 Conclusions

The Bayesian renewal equation with GP residual provides a principled framework for decomposing dengue R_t into climate-attributable and residual components. Climate is genuinely negligible at weekly resolution. The residual GP captures multi-scale temporal structure descriptively consistent with serotype dynamics — a ~ 3.6-year peak consistent with the multi-annual oscillations transient cross-immunity can generate in total dengue incidence [Wearing and Rohani, 2006], 5 of 7 serotype switches associated with elevated R_t, and serotype turnover leading residual R_t elevation by approximately 8 months. These associations are descriptive rather than mechanistic: a separate model-comparison audit across a susceptible-fraction sensitivity sweep shows that explicit $\log S(t)$ covariates (aggregate or serotype-aware) do not absorb the GP residual, so the residual is tracking something that simple SIR pool accounting cannot represent. The framework also has fundamental limitations for operational forecasting — the GP provides no predictive skill beyond persistence because it is a smoothing prior, not a forward dynamical model. The cross-kernel comparison confirms robustness across four standard kernels. Future work should integrate GP decomposition with spatially explicit, multi-serotype immunity models that can support both causal attribution and forward simulation.

Code Availability

All code and data for this project are available as a zipped repository submitted alongside this report. The repository includes:

- **Data acquisition:** Python scripts using `requests` and `meteostat` to download dengue cases (data.gov.sg), weather (Meteostat API), and serotype proportions (NEA).
- **Model specification:** Stan code (`05_model_climate_only.stan`) implementing the HSGP renewal equation model.

- **Analysis pipeline:** R scripts numbered in execution order for data preparation (`03_prepare_model_data.R`), prior predictive checks (`04_prior_predictive.R`), model fitting (`06_fit_model.R`), posterior predictive checks (`07_posterior_predictive.R`), posterior summaries and variance decomposition (`08_postprocess.R`), retrospective residual decomposition (`09_residual_decomposition.R`), serotype analysis (`10_serotype_analysis.R`), climate-lag and susceptible sensitivity analyses (`12_lag_sensitivity.R`, `13_susceptible_sensitivity.R`), MCMC convergence diagnostics (`14_convergence_diagnostics.R`), and GP hyper-prior sensitivity (`15_gp_prior_sensitivity.R`). The susceptible-covariate audit discussed in Section 4.3 lives in `14_prepare_susceptible_covariates.R`, `14_model_climate_susceptible.stan`, `14_fit_susceptible_models.R`, and `14_compare_susceptible_models.R`. Extended analyses — kernel comparison, forecasting evaluation, branching process analysis, and spectral analysis — are in `code/archive/`. A preliminary VAE pilot is also archived but not reported in the main results.
- **Report:** Quarto source (`report.qmd`) and compiled PDF.

Scripts are numbered in execution order. A `README.md` provides setup instructions. R dependencies are managed via `renv`; Python dependencies via `uv`. The Stan model requires CmdStan $\geq$ 2.38.0.

Declaration on Use of AI Tools

Claude (Anthropic) was used as a coding assistant for Stan model development, R/Python analysis scripts, and to assist with drafting and editing this report. All scientific decisions, model specification choices, and interpretation of results were made by the author. Code and analysis were reviewed and validated by the author.

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